Adaptation of Mammalian Cells to Chemically Defined Media

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In order to circumvent ethical, technical, and economic drawbacks regarding the use of animal serum in cell culturing, it is possible to adapt mammalian cells to serum-free media. Nowadays, there are several serum-free formulations available, including fully animal derived–free and chemically defined media, and different adaptation techniques. This article focuses on the gradual adaptation of a mammalian suspension cell culture to a chemically defined medium. The first step is to transfer the cells cultured in medium supplemented with fetal bovine serum (FBS) to a chemically defined medium of your choice, containing the same amount of FBS. The next steps consist of progressively reducing the amount of FBS, while monitoring cell growth and viability up to the complete elimination of FBS. This protocol has been successfully used to adapt THP-1 cells to a chemically defined medium with similar maximum specific growth rate as those cultured with FBS. © 2019 by John Wiley & Sons, Inc.

Basic Protocol: Gradual adaptation to chemically defined medium
Alternate Protocol: Direct adaptation to chemically defined medium
Support Protocol 1: Determining maximum specific growth rate of a cell culture
Support Protocol 2: Cell freezing and thawing

Keywords: animal serum free • animal use alternatives • cell culture techniques • culture media • serum-free

How to cite this article:
Marigliani, B., Balottin, L. B. L., & Augusto, E. F. P. (2019). Adaptation of mammalian cells to chemically defined media. *Current Protocols in Toxicology*, *82,* e88. doi: 10.1002/cptx.88

INTRODUCTION

Despite animal welfare concerns and several technical and economic drawbacks regarding the use of fetal bovine serum (FBS) for cell culture supplementation, this is still a standard practice (van der Valk et al., 2018). The remarkable increase in the availability of animal serum–free medium alternatives—such as human serum, human serum albumin, human platelet lysate, and chemically defined medium (Karnieli et al., 2017; van der Valk et al., 2010, 2018)—along with a number of recommendations for FBS replacement (Coecke et al., 2005; Geraghty et al., 2014; Organisation for Economic Co-operation and Development, 2018) leads to a practical issue: how to replace FBS in culture medium.

CURRENT PROTOCOLS A Wiley Brand

Current Protocols in Toxicology e88, Volume 82
Published in Wiley Online Library (wileyonlinelibrary.com).
doi: 10.1002/cptx.88

First, a suitable serum-free medium must be chosen. Several serum-free formulations have already been described for mammalian cells, and there are a number of commercially available serum-free media (Brunner et al., 2010). In order to circumvent the disadvantages of using animal serum or human blood derivatives—such as ethical concerns, lot-to-lot variability, and the risk of contamination (van der Valk et al., 2018)—a chemically defined medium (or more than one) must be chosen. This can be done by searching the literature and media manufacturer catalogues, which can be time consuming, but a freely accessible interactive online database of serum-free media and adapted cell lines is available (Brunner et al., 2010). Choosing a culture medium that has already been used to adapt a cell line that is similar to the medium of interest (e.g., same type of cell) is a good start, but adaptation is often a trial-and-error process.

In addition to the variety of media available—including serum-free, protein-free, animal derived–free, and chemically defined media—several approaches to adapt cells to these media have proven successful for different cells. Thus, after choosing one or more chemically defined medium, the next step is to select an adaptation scheme to follow. One of these is gradual adaptation, a weaning process in which the percentage of FBS is gradually reduced in the culture medium (van der Valk et al., 2010).

If adaptation is successful, the next step is to create a serum-free cell bank and to ensure that the adapted cells are equivalent to the ones cultured in FBS-containing medium. The latter can be done by assessing if the adapted cells provide the same results in chosen experiments (e.g., by using them in validated protocols, such as what was done with THP-1 cells; Marigliani et al., 2019) or by authentication (e.g., by performing short tandem repeat [STR] DNA profiling; Reid, Storts, Riss, & Minor, 2013). This is an important step to guarantee that the adaptation process does not modify the cell model in an extent that could compromise future work.

Here we describe a variation of a protocol to gradually adapt mammalian cells to serum-free medium (Sinacore, Drapeau, & Adamson, 2000). The first step of the original protocol was the adaptation to suspension culture, as here we use cells that grow in suspension. This first step, however, is not necessary, and we can go straight for the adaptation to serum-free medium. This modified protocol was successfully used to adapt THP-1 cells—a human monocytic cell line derived from a patient with acute monocytic leukemia (Tsuchiya et al., 1980), available from the American Type Culture Collection (ATCC; #TIB-202)—to a chemically defined medium (Marigliani et al., 2019).

STRATEGIC PLANNING

All adaptation processes should start with cells from an exponential growth phase (see Support Protocol 1), with >90% cell viability. The use of antibiotics is not recommended, and regular testing should be done to detect cell culture contamination. However, some chemically defined media contain antibiotics, such as gentamicin, which are present in the medium used in this protocol (e.g., X-VIVO 10, Lonza, cat. no. 04-380Q).

It is important to assess the maximum specific growth rate (μ_{Max}) of cells cultured in FBS-containing medium before starting the adaptation process. See Support Protocol 1 for details on determining μ_{Max} and for its relation to doubling time (DT), which is a reference value for cells cultured in FBS-containing medium. Although historical data can be obtained from the cell bank provider, we strongly suggest building an internal database. For gradual adaptation, after assessing μ_{Max} for cells cultured in FBS-containing medium, cells are transferred to the chemically defined medium with the same percentage of FBS. Then this FBS percentage is gradually reduced, up to full FBS elimination. Cell viability and μ_{Max} should be assessed during the entire adaptation process and at the end to confirm cell adaptation.

GRADUAL ADAPTATION TO CHEMICALLY DEFINED MEDIUM

BASIC PROTOCOL

Elimination of FBS during cell culturing can be performed in a single-step approach (see the Alternate Protocol), but gradual adaptation generally works better. In this technique, the amount of FBS in culture medium is reduced in a stepwise way, and cells can adapt to the new condition (i.e., reduced amount of FBS) before performing the next FBS reduction.

This protocol provides detailed steps for the adaptation of a mammalian cell line (THP-1), which is originally cultured in FBS-containing medium (10% [v/v]), to a chemically defined medium. The procedures described can be used to adapt other animal cell lines to other FBS-free media.

In order to avoid losing the culture and having to start the procedure over again, at each FBS reduction step, it is important to keep a backup culture with the previous amount of FBS and to only discard this backup culture when cells at the next reduction step are adapted. It is also important to determine the expected DT (or μ_{Max}) for cells cultured in usual medium (i.e., with the recommended amount of FBS, according to the cell bank) and for your specific laboratory conditions, so that these parameters can be compared to the ones obtained for cells during the adaptation process (see Support Protocol 1). Creating a cell bank of adapted cells and storing backup cultures during the adaptation process is very important, and therefore a freezing protocol is also provided (see Support Protocol 2).

Materials

Mammalian cells (e.g., THP-1 cells, ATCC #TIB-202; RRID:CVCL_0006) at exponential growth phase (see Support Protocol 1)
Usual cell culture medium appropriate for cells of interest (see recipe)
Chemically defined medium (see recipe)
FBS

25-cm^2 cell culture flasks, untreated (e.g., Corning, cat. no. 431463)
37°C, 5% CO_2 cell culture incubator
Hemacytometer or other method to determine cell concentration
15-ml conical tubes
Refrigerated centrifuge

Grow cells in usual medium

1. Grow mammalian cells in 25-cm^2 flasks in usual medium, according to the instructions of the cell bank provider or to your standard laboratory protocol, and determine μ_{Max} for this condition (see Support Protocol 1).

Transfer cells to chemically defined medium supplemented with FBS

2. To start the adaptation protocol, obtain cells cultured in usual medium at the exponential growth phase and at the lowest passage number possible (e.g., after thawing, as soon as μ_{Max} becomes constant, which means that cells have recovered from freezing and thawing processes).

3. Determine cell concentration (cells/ml), and in 15-ml conical tubes, centrifuge cell suspension volume necessary to obtain 1.3×10^6 cells for 5 min at $250 \times g$, 4°C.

4. Resuspend cell pellet in 1.3 ml conditioned medium plus 5.2 ml chemically defined medium containing 10% (v/v) FBS in order to obtain 0.2×10^6 cells/ml in 6.5 ml final volume (i.e., 20% conditioned medium and 80% fresh medium).

   For this and subsequent steps, conditioned medium indicates the usual or chemically defined medium in which cells were grown.

5. Transfer 6.0 ml cell suspension to a 25-cm^2 flask, and incubate 3 days (time necessary for achieving exponential growth phase, which can vary depending on the cell lineage). Use the remaining 0.5 ml to determine the inoculum cell concentration and viability.

Remove usual medium

6. Determine cell concentration, and calculate volume necessary to obtain 1.3×10^6 cells.
7. Centrifuge cells 5 min at $250 \times g$, 4°C, and resuspend cell pellet in 6.5 ml fresh medium (chemically defined medium with 10% FBS).
8. Transfer 6.0 ml cell suspension to a new 25-cm^2 flask in order to obtain 0.2×10^6 cells/ml in 6.0 ml (100% fresh medium). Incubate cells 2 to 3 days (time necessary for achieving exponential growth phase, which can vary depending on the cell lineage). Use the remaining 0.5 ml to determine the inoculum cell concentration and viability.

FBS weaning process

Once cells are cultured in chemically defined medium supplemented with FBS, you can start the FBS weaning process. This is done by gradually reducing the percentage of FBS when cells are clearly adapted to the current condition (see Support Protocol 1). The percentages of FBS in the reduction steps used here are: 5.0, 2.5, 1.25, 0.625, 0.312, and 0% (v/v) FBS.

After each FBS reduction step, determine cell concentration, and assess cell viability and μ_{Max} every 2 to 3 days. Then subculture cells using 1.3×10^6 cells in a final volume of 6.5 ml. Use 1.3 ml conditioned medium plus 5.2 ml fresh medium with the same percentage of FBS during adaptation; or, when cells are adapted to the current passage (next reduction step), use 5.2 ml fresh medium supplemented with the next lower percentage of FBS. Always transfer 6.0 ml to the 25-cm^2 flask, and use the remaining 0.5 ml to determine the inoculum cell concentration and viability. Here, we consider adapted cells the ones that present >90% cell viability and a μ_{Max} that is similar to (or higher than) the μ_{Max} observed for cells cultured in the usual medium (e.g., 10% [v/v] FBS).

Continue reducing the percentage of FBS up to full elimination (0% FBS) by passing cells two to three times a week, using 20% (v/v) conditioned medium and 80% (v/v) fresh medium, except for the first passage after each reduction step, when only fresh chemically defined medium must be used (see steps 6 to 8). When cells are adapted to the 0% FBS condition, make a cryopreserved cell bank (see Support Protocol 2).

As previously mentioned, always keep a backup cell culture at each FBS reduction step, and only discard the backup culture when cells at a lower FBS reduction step are adapted to that condition. If, at any FBS reduction step, cells are taking too long (>10 passages) to adapt (identified by a sharp fall in cell viability without recovery or failure to present similar or higher μ_{Max} than the previous condition), discard the cells, and use the backup cell culture (previous percentage of FBS) to perform another reduction attempt, which can be done with the exact same reduction or with a less drastic one. For example, if the reduction from 5% to 2.5% FBS does not work, try reducing from 5% to 4.0% or to 3.5% and so on.

DIRECT ADAPTATION TO CHEMICALLY DEFINED MEDIUM

ALTERNATE PROTOCOL

In the direct adaptation protocol, cells cultured in FBS-containing medium are adapted in a one-step approach, without a weaning process. Although less effective, direct adaptation can be a first attempt, as it is less demanding and time consuming.

Materials

Mammalian cells (e.g., THP-1 cells, ATCC #TIB-202; RRID:CVCL_0006) at exponential growth phase (see Support Protocol 1)
Usual cell culture medium appropriate for cells of interest (see recipe)
Chemically defined medium (see recipe)
FBS

25-cm^2 cell culture flasks, untreated (e.g., Corning, cat. no. 431463)
37°C, 5% CO_2 cell culture incubator
Hemacytometer or other method to determine cell concentration
15-ml conical tubes
Refrigerated centrifuge

Grow cells in usual medium

1. Grow mammalian cells in 25-cm^2 flasks in usual medium according to the instructions of the cell bank, and determine μ_{Max} (see Support Protocol 1).

Transfer cells to chemically defined medium

2. Determine cell concentration, and in 15-ml conical tubes centrifuge cell suspension volume necessary to obtain 1.3×10^6 cells for 5 min at 250 × *g*, 4°C.

3. Resuspend cell pellet in 1.3 ml conditioned medium plus 5.2 ml chemically defined medium (without FBS supplementation), in order to obtain 0.2×10^6 cells/ml in 6.5 ml final volume (i.e., 20% conditioned medium and 80% fresh medium).

   For this and subsequent steps, conditioned medium indicates the usual or chemically defined medium in which cells were grown.

4. Transfer 6.0 ml cell suspension to a 25-cm^2 flask, and incubate for 3 days (time necessary for achieving exponential growth phase, which can vary depending on the cell lineage). Use the remaining 0.5 ml to determine the inoculum cell concentration and viability.

Remove usual medium

5. Determine cell concentration and viability, and calculate volume necessary to obtain 1.3×10^6 cells.

6. Centrifuge cells 5 min at 250 × *g*, 4°C, and resuspend cell pellet in 6.5 ml fresh medium (chemically defined medium without FBS) in order to obtain 0.2×10^6 cells/ml in 6.5 ml (100% fresh medium).

7. Transfer 6.0 ml cell suspension to a 25-cm^2 flask, and incubate 2 to 3 days (time necessary for achieving exponential growth phase, which can vary depending on the cell lineage). Use the remaining 0.5 ml to determine the inoculum cell concentration and viability.

Adaptation

8. Subculture cells every 2 to 4 days (time necessary for achieving exponential growth phase, which can vary depending on the cell lineage) using 1.3 ml conditioned medium plus 5.2 ml chemically defined medium (0% [v/v] FBS), starting with

0.2×10^6 cells/ml in 6.5 ml final volume (i.e., 20% conditioned medium and 80% fresh medium). Monitor cell concentration, viability, and μ_{Max}/DT.

Alternatively, starting the culture with 0.4×10^6 cells/ml is also possible if starting with 0.2×10^6 cells/ml does not provide good cell growth and viability.

9. Culture cells in this condition until adaptation is achieved, and prepare a cryopreserved cell bank (see Support Protocol 2).

Adapted cells are those that present >90% cell viability and a μ_{Max} that is similar to (or higher than) the μ_{Max} observed for cells cultured in the usual medium (e.g., 10% [v/v] FBS).

SUPPORT PROTOCOL 1

DETERMINING MAXIMUM SPECIFIC GROWTH RATE OF A CELL CULTURE

This protocol describes the steps involved in assessing μ_{Max} of a cell culture. The μ_{Max} is proposed as a tool to compare cells during the adaptation process to those cultured in FBS-containing medium, as well as to characterize adapted cells.

Materials

Cells at desired growth phase (see Basic Protocol or Alternate Protocol)
0.4% (v/v) trypan blue
Hemacytometer (e.g., Neubauer)
Data analysis software (e.g., Microsoft Excel)

1. Determine maximum specific growth rate: At any time during cell culture, determine cell concentration and viability using a hemacytometer and trypan blue staining (see Freshney, 2010), and calculate the specific growth rate (μ), which can be defined as the cellular growth rate (r_X) divided by the cellular concentration (X), as shown in Equation 1:

$$\mu = \frac{1}{X} \cdot r_X = \frac{1}{X} \cdot \frac{dX}{dt}$$

Equation 1

where,

μ = specific growth rate (1/hr),
X = cell concentration at t (cells/ml),
r_X = growth rate (cells/ml/hr), and
t = time (hr).

Specific growth rate represents the behavior of each cell in the system and, therefore, puts everything on a comparable basis; thus, it is the best parameter for kinetic analysis and for assessing the physiologic state of cells (Gaden, 2000).

Equation (1) can be easily integrated for the period of time corresponding to the exponential growing phase, when μ assumes a constant and maximum value (μ_{Max}), resulting in Equation 2:

$$\ln(X) = \ln(X_{in}) + \mu_{Max} \cdot (t - t_{in})$$

Equation 2

where,

μ_{Max} = maximum specific growth rate (1/hr),

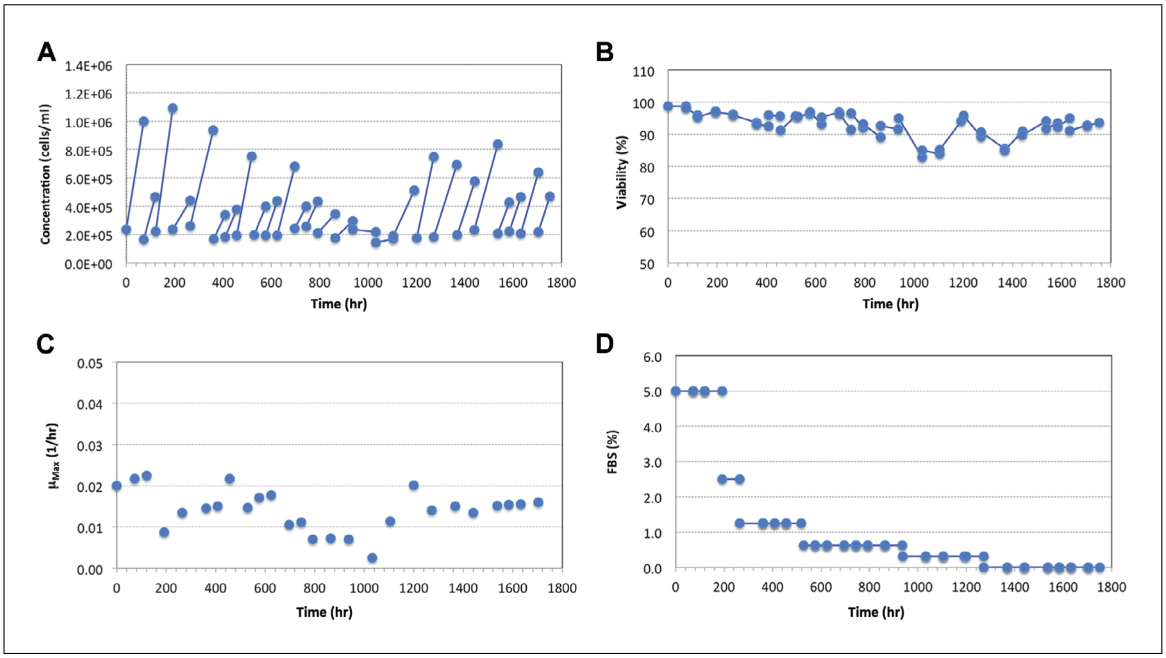

Figure 1 Example of successful gradual adaptation to chemically define medium. (**A**) Cell concentration, (**B**) cell viability, (**C**) maximum specific growth rate (μ_{Max}), and (**D**) percentage of fetal bovine serum (FBS) in the culture medium.

X = cell concentration at t (cells/ml),
X_{in} = cell concentration at the initial time of exponential growing phase (cells/ml/hr),
t = time during the exponential growing phase (e.g., the final time of subculturing; hr), and
t_{in} = initial time for the exponential growing phase (e.g., innoculum time of the new subculture; hr).

Thus, on a plot of the natural logarithm of cellular concentrations per time, the exponential growing phase can be identified as the period of time with a linear correlation between the variables, as indicated in Equation 2. The μ_{Max} can be determined as the slope of the equation.

2. Apply maximum specific growth rate to evaluate adaptation process.

During the process of cell adaptation to FBS-free medium, subculturing is preferably done on exponential phase; thus, it is possible to apply Equation 2 to data (cell concentration) obtained at the beginning and at the end of culturing and to determine μ_{Max} for the specific passage. The behavior of μ_{Max} along this process is used to validate cell adaptation to the new culture condition: it will become approximately constant. This can be similar to or higher than the μ_{Max} for the same cells cultured in FBS-containing medium.

Figure 1 shows an example of a successful adaptation process. Notice that there is a clear trend toward a consistent μ_{Max} in the last seven passages.

3. Monitor DT to evaluate the adaptation process.

Equation 3 gives the relationship between μ_{Max} and DT, the latter of which can also be used as an adaptation indicator. DT will be similar to or lower than cells grown in FBS-containing medium.

$$DT = \frac{\ln(2)}{\mu_{Max}}$$

Equation 3

SUPPORT PROTOCOL 2

CELL FREEZING AND THAWING

It is of utmost importance to make a cell bank of adapted cells, but it is also recommended to cryopreserve cells during the adaptation process in order to have backup cells adapted to different FBS percentages to overcome any experimental disturbance. The freezing medium is usually composed of the complete growth medium (e.g., RPMI-1640 medium supplemented with 10% FBS) and a cryoprotective agent (e.g., 5% dimethyl sulfoxide [DMSO]), which is added to prevent the formation of ice crystals.

In order to completely eliminate the use of FBS, besides adapting cells to a chemically defined medium, it is also necessary that the adapted cells be cryopreserved without FBS. This protocol describes the freezing and thawing process of cells without FBS.

Materials

Cells to be frozen (see Basic Protocol or Alternate Protocol)
Chemically defined medium (see recipe)
Chemically defined freezing medium (see recipe)
100% isopropyl alcohol

Hemacytometer
Refrigerated centrifuge
2.0-ml cryogenic vials with internal thread (e.g., Corning, cat. no. 430488)
Freezing container (e.g., Mr. Frosty)
Liquid nitrogen tank
37°C water bath
15-ml conical tubes
25-cm^2 cell culture flasks, untreated (e.g., Corning, cat. no. 431463)

Prepare cells for freezing

1. For cell freezing, prepare cells in exponential growth phase, and determine cell concentration.

2. Calculate the volume necessary to obtain enough cells to have 6×10^6 cells in each vial, and centrifuge this volume 5 min at $250 \times g$, 4°C.

Add freezing medium

3. Resuspend cell pellet in 0.5 ml precooled (2°C to 8°C) chemically defined medium per vial and keep chilled.

4. Slowly add an equal volume chemically defined freezing medium (1 ml total volume), and gently pipette up and down to homogenize cell suspension.

   The final concentration of DMSO will be 7.5%, and the final concentration of cells will be 6×10^6 cells/ml.

Freeze vials

5. Transfer 1 ml cell suspension to a 2-ml cryogenic vial, and then transfer vials to a freezing container filled with 100% isopropyl alcohol. Store container at −80°C, and after 24 to 48 hr, place vials into vapor phase liquid nitrogen.

Thaw cells

6. For thawing cells, remove vial from the liquid nitrogen tank. In a sterile field, briefly open the cap by twisting it a quarter of a rotation, and close it again.

7. Quickly thaw vial in a 37°C water bath without submerging it.

8. Gently pipette up and down to resuspend cells, and transfer to a 15-ml conical tube. Slowly add 9 ml prewarmed chemically defined medium.

Start new batch

9. Determine cell concentration and viability, and then adjust volume to have 1.3×10^6 cells in 6.5 ml chemically defined medium (0.2×10^6 cells/ml).

10. Transfer cell suspension to a new 25-cm^2 flask, and incubate cells for 3 days.

REAGENTS AND SOLUTIONS

Chemically defined freezing medium

Supplement non-animal origin freezing medium (e.g., ProFreeze-CDM, Lonza, cat. no. 12-769E) with 15% (v/v) DMSO (e.g., Sigma-Aldrich, cat. no. D2438). Store on ice, and use within 8 hr.

Chemically defined medium

Supplement X-VIVO 10 serum-free cell culture medium (e.g., Lonza, cat. no. 04-380Q) with 2-mercaptoethanol (e.g., Thermo Fisher Scientific, cat. no. 21985023) to a final concentration of 0.05 mM. Store at 4°C protected from light for up to 2 months.

Usual medium

Supplement RPMI-1640 medium (e.g., Lonza, cat. no. 12-115Q) with 10% FBS (e.g., Thermo Fisher Scientific, cat. no. 10437028) and 2-mercaptoethanol (e.g., Thermo Fisher Scientific, cat. no. 21985023) to a final concentration of 0.05 mM. Store at 4°C for up to 2 months.

COMMENTARY

Background Information

The adaptation of mammalian cells to serum-free medium can be tricky and time consuming. The protocols described explain the steps necessary to conduct gradual and direct adaptation of cells to a chemically defined medium.

Other successful adaptation strategies have already been described, such as a three-step protocol for the adaptation of mammalian cells (Chinese hamster ovary [CHO] cells) to serum-free media, a synthetic surfactant that increases cell viability in serum-free medium after thawing, adaptation approaches for anchorage-dependent mammalian cells to a suspension serum-free culture condition, adaptation of CHO cells to serum- and protein-free conditions for recombinant protein production, gradual adaptation of a fish cell line to serum-free conditions in 32 days, and adaptation of Vero cells to suspension culture using different serum-free and chemically defined media (Caron, Biaggio, & Swiech, 2018; González Hernández & Fischer, 2007; Jukić, Bubenik, Pavlović, Tušek, & Srček, 2016; Radošević, Dukić, Andlar, Slivac, & Gaurina Srček, 2016; Rourou, Ben Zakkour, & Kallel, 2019; Sinacore et al., 2000).

The described protocols were developed for suspended cell lines. For plated cells, if having suspension cells as a final result is acceptable, the adaptation to suspension culture should be done before beginning these protocols (see Sinacore et al., 2000). For primary culture cells, the ideal scenario is to start the culture using a serum-free medium, so that the adaptation process is not needed.

Critical Parameters

The maximum time for cell propagation after thawing, as well as the maximum number of passages and cell density as recommended by the cell bank provider for cells cultured in FBS-containing medium, should also be applied for adapted cells. However, as the adaptation process can exceed the recommended number of passages and/or time after thawing, it is important to characterize the adapted cells, which can be done by comparing μ_{Max}, an easy parameter to routinely assess in the laboratory. However, the comparison of results provided by adapted cells and cells cultured in FBS-containing medium in chosen experiments, as well as authentication by STR DNA profiling, is also recommended.

Care should be taken to avoid contamination, as the use of antibiotics is not recommended during the adaptation process.

FBS and other animal serum have a concentration-dependent protective effect on animal cells. The precise nature of the physical protective effect of serum is not clear, and there are some animal (e.g., bovine serum albumin) and non-animal alternatives to provide shear protection for mammalian cells, such as Pluronic F68 and polyethylene glycol (Chisti, 2000). The non-ionic surfactant Pluronic F-68 can be added to cultures of adapted cells in order to form a protective layer on the cell membrane, which prevents loss of cell viability and reduces cell adhesion (Tharmalingam, Ghebeh, Wuerz, & Butler, 2008). As the cell membrane of mammalian cells can be sensitive to shear stress, the cell suspension must be carefully suspended by gentle pipetting, in order to avoid shear stress bubble-associated damage during the adaptation process.

Troubleshooting

During the adaptation process or immediately after thawing adapted cells, cells may attach to each other and form clumps. These aggregates reduce cell access to nutrients and may reduce cell growth. Cell clumps can be dispersed by using an animal origin–free chemically defined anti-clumping agent (e.g., Gibco, cat. no. 0010057AE).

A couple of strategies can be applied if cells present a sharp drop in viability or μ_{Max} immediately after an FBS-reduction step: (1) Use a less drastic FBS reduction step (e.g., from 2.5% to 1.75% FBS if 2.50% to 1.25% is too drastic). (2) Increase the cell concentration (e.g., using 0.4×10^6 cells/ml instead of 0.2×10^6 cells/ml) at the reduction step.

Adapting cells must be constantly monitored for biological contamination (bacteria, yeast, fungi, mycoplasma). If any contamination is detected, it is recommended to discard the contaminated cells and all media formulations used with these cells and to use the backup culture (provided it is free of contamination), thaw a new lot of cells, or restart the adaptation process.

If, at any step during gradual adaptation, the cells surpass the maximum recommended cell density for cells cultured in FBS-containing medium (e.g., 1×10^6 cells/ml for THP-1 cells) or enter stationary growth phase, discard these cells, and use a backup cell culture. If this happens when cells cultured in FBS-containing medium are transferred to chemically defined medium supplemented with the same percentage of FBS (e.g., 10% FBS), transfer cells cultured in usual medium (e.g., supplemented with 10% FBS) to a chemically defined medium with a lower FBS percentage (e.g., 5% FBS). Alternatively, incubate cells with the same FBS percentage for a shorter period of time to avoid reaching the stationary growth phase.

Understanding Results

The constant μ_{Max} (which is similar to or higher than μ_{Max} for cells cultured in FBS-containing medium) is a good and fast indicator that cells are adapted to chemically defined medium. However, it is important to assess if the adapted cells can be frozen and thawed for a number of cycles without reductions in cell viability and μ_{Max}, as well as if they keep these parameters constant throughout long-term cell cultures. Also, as previously mentioned, adapted cells must be tested by using them in protocols developed with cells cultured in FBS-containing medium, such as what has been done with THP-1 cells (Marigliani et al., 2019).

Time Considerations

Assuming cells cultured in FBS-containing medium at the exponential growth rate are available, gradual adaptation can be performed in ~6 weeks, considering one reduction step every week. Here we describe a process that takes around 10 weeks, taking into account the last seven passages without FBS. Direct adaptation is faster, as it takes a single step, and cells are considered to be adapted as soon as they present >90% cell viability and a μ_{Max} that is similar to (or higher than) the μ_{Max} observed for cells cultured in medium with FBS, which can be achieved in 2 weeks.

Acknowledgments

This work was supported by the 2015 Lush Prize (to B.M., young researcher category), Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES, Brazil), and a PRONAMETRO fellowship. The authors are grateful to the National Institute of Metrology, Quality and Technology (INMETRO), where the experiments for this protocol were performed.

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Key Reference

Freshney (2010). See above.

Cell counting procedures described within this article provided in chapter 20.

Internet Resource

http://www.sefrec.com

Interactive database with information about serum-free media and cell lines.